

ORIGINAL ARTICLE

Combined treatment with triptolide and rapamycin prolongs graft survival in a mouse model of cardiac transplantation

Yan Liu,^{1*} Yan Chen,^{1*} Fing Qin Liu,¹ Jonathan R. Lamb² and Paul K. H. Tam¹

¹ Division of Paediatric Surgery, Department of Surgery, University of Hong Kong Medical Centre, Queen Mary Hospital, Hong Kong, China

² Department of Veterinary Clinical Studies, Royal (Dick) School of Veterinary Studies, University of Edinburgh, Easter Bush, Roslin, Midlothian, UK

Keywords

cardiac allografts, CD4⁺CD25⁺Foxp3⁺ T cells, DC-SIGN, rapamycin, Th1/Th2 cytokines, triptolide.

Correspondence

Professor Paul K. H. Tam, Department of Surgery, The University of Hong Kong, Queen Mary Hospital, Pokfulam, Hong Kong, China. Tel.: 852 28554850; fax: 852 28173155; e-mail: paultam@hkucc.hku.hk

*These two authors contributed equally to this work.

Received: 21 September 2007

Revision requested: 24 October 2007

Accepted: 12 December 2007

doi:10.1111/j.1432-2277.2007.00630.x

Summary

Current immunosuppressive strategies for transplantation have failed to achieve long-term graft survival. In this study, we investigate the effects of combined treatment with triptolide (TPT) and rapamycin (Rapa) on graft survival as well as changes in pathology and immunological responses. Heterotopic heart transplantation was performed. TPT and Rapa were administered either alone or in combination. The mean survival time (MST) for the cardiac allografts in animals receiving the combination of TPT and Rapa was 93.5 ± 6.7 days compared to treatment with TPT (MST: 23.5 ± 5.3 days), Rapa (22 ± 1.3 days) alone or no treatment (7.66 ± 0.8 days). Histopathological evaluation showed that inflammatory cell infiltration was markedly reduced in grafts with combined treatment groups. Down-regulation of CCL19, CCR5, CCR7, interferon γ and interleukin (IL)-12 in the combination treatment was accompanied by increased expression of IL-4, IL-10 and CD4⁺CD25⁺Foxp3⁺ regulatory T (Tr) cells in spleen. Finally, dendritic cell (DC) maturation was impaired by treatment with TPT/Rapa. Our results demonstrate that combination therapy with TPT and Rapa markedly prolongs cardiac allograft survival. This effect is accompanied by inhibition of DCs maturation, conditioning DCs to adopt tolerogenic phenotype, and the expansion of Tr cells. These results add weight to the application of combination therapy in transplantation.

Introduction

Current immunosuppressive drugs have dramatically improved the management of acute graft rejection (AR) but fail to control chronic rejection (CR). Indeed when administered at high doses, some compounds promote CR, for example cyclosporine A [1]. Therefore, the development of improved immunosuppressive regimens was needed. Improvements were made, and regimens are now available that consist of immunosuppressive agents that target different parts of the immune response to the allograft and, therefore, work well in combination.

Triptolide (TPT), a major active component of the Chinese herb *Tripterygium wifordii* Hook F (TWHF), is a traditional Chinese medicine and has been used for

the treatment of autoimmune diseases such as rheumatoid arthritis and systemic lupus erythematosus for many years [2,3]. In the past decade, the clinical applications of TWHF and TPT have been extended from autoimmune diseases to organ transplantation [4,5]. Previous studies have shown that TPT possesses potent immunosuppressive and anti-inflammatory properties [5]. TPT also has inhibitory effects on the maturation and allostimulation of human monocyte-derived dendritic cells (DCs) [6]. The immunosuppressive activity of rapamycin (Rapa) in organ transplantation has been studied in detail *in vivo* [7], and *in vitro* it has been demonstrated to inhibit alloantigen-induced T-cell proliferation, DC maturation and the secretion of inflammatory cytokines [8].

Previous studies have investigated the combined effects of TPT together with FK506 [9] and cyclosporine A [10] on graft survival. The mode of action of both of FK506 and cyclosporine A is through inhibition of calcineurin-NFAT signal transduction in lymphocytes and which results in the suppression of AR [11]. In this study, we have examined the effects of TPT combined with Rapa (TPT/Rapa) on cardiac graft rejection. Rapa binds to FKBP12, as does FK506, but also inhibits the function of the protein kinase mammalian target of rapamycin (mTOR). Our results demonstrate that TPT/Rapa can significantly prolong allograft survival in a fully MHC-mismatched cardiac transplant model in mice. In addition to reducing the inflammatory cell infiltrate, TPT/Rapa treatment also results in the expansion of a population of CD4⁺CD25⁺Foxp3⁺ regulatory T (Tr) cells. Furthermore, the combination therapy modulated chemokine and cytokine expression in favour of a homeostatic phenotype. We also demonstrate the different mechanisms of these two immunosuppressive drugs on DC function. While both compounds can inhibit DC maturation and reduce Toll-like receptor (TLR4) expression, only TPT increased the expression of DC-SIGN. These findings provide insight into the potential clinical application of combination therapies using immunosuppressive drugs in the management of graft rejection.

Materials and methods

Mice and reagents

Inbred male C57Bl/6 (B6, H-2^b), BALB/c (H-2^d), and 129x1/SvJ (129, H2) mice (6–8 weeks old) were purchased from the Animal Resources Centre, Australia. Animals were allowed free access to food and water in a 12-h light, 12-h dark cycled room. The experimental protocols were approved by the Committee on the Use of Live Animals in Teaching and Research, The University of Hong Kong.

Triptolide [TPT, C₂₀H₂₄O₆, molecular weight (MW) 360, Purity (HPLC) ≥ 98%] was purchased from Sigma (Sigma-Aldrich, St. Louis, MO, USA) and was reconstituted in dimethyl sulfoxide (DMSO) at concentration of 10 mM and was stored in –20 °C. The working solution of TPT was freshly prepared by dilution of TPT in culture medium. Rapa (1 mg/ml) was purchased from Wyeth-Ayerst (Ayerst Laboratories, Pearl River, NY, USA) and diluted in saline for doses of 0.1 mg/ml.

Cardiac transplantation model and treatment protocols

Heterotopic cardiac transplantation was performed as previously described [12]. Cardiac allografts from B6 mice

were transplanted heterotopically into BALB/c mice. Vascularized grafts were transplanted according to standard microsurgical techniques. Briefly, donor ascending aorta and the pulmonary trunk from the heart graft was anastomosed end-to-side to the recipient's abdominal aorta and inferior vena cava, respectively. The cold ischaemia period lasted <20 min. In this experimental model, AR usually occurs between days 5 and 7 after transplantation [8]. No other supportive measures were required during the surgery. Acute cardiac rejection was defined by cessation of a palpable heart beat and was confirmed by histology.

The doses of the TPT were chosen on the basis of the results of our previous experiments, in which we observed that administration of 3 mg/kg per day of TPT caused significant prolongation of survival of cardiac allografts in mice [13]. The dose of Rapa (1 mg/kg/day) used was as previously published [14] and administered orally. Both of these immunosuppressants were given at days 0–7, 9, 11, 13 and 15 post-transplantation.

Histological and immunohistochemical analysis

The hearts (transplanted and native) were removed and divided in half in long axis perpendicular to the intraventricular septum. The formalin-fixed tissue was embedded in paraffin, and 4-μm sections were cut and stained with haematoxylin and eosin (H&E) for histology examination. Immunohistochemical staining was performed using the same sections as described above. The endogenous peroxidase was quenched by treated-in 3% hydrogen peroxide/methanol for 10 min. For antigen retrieval for T-cell staining, slides were treated with Tris-EDTA Buffer (pH 9.0) at 95–100 °C for 20 min. For macrophage and DC staining, the sections were treated with proteinase K solution in humidified chamber for 20 min at 37 °C. The sections were further blocked for nonspecific binding by normal goat serum (1/100) and CD16/CD32 antibody (Ab) before the primary Ab was added. CD3 Ab (Serotec MCA1477; Serotec, Kidlington, Oxford, UK), F4/80 Ab (Serotec MCA497) and 33D1 Ab (ebioscience 14-5884) were used with 1/100 dilution and incubated overnight at 4 °C. As a control for background staining of the primary Ab an IgG isotype was used. Sections were then rinsed in PBS and incubated with HRP-conjugated goat anti-rat IgG secondary Ab (Santa Cruz Biotechnology, Santa Cruz Inc., CA, USA) for 1 h. The positive signals were developed using DAB (3,3'-diaminobenzide tetrahydrochloride) and counterstained with haematoxylin. The degree of inflammatory cell infiltration was evaluated by counting and averaging total nuclear number in 10 randomly selected fields by microscopy (400×).

Analysis of tissue expression of chemokine receptors/ligands and cytokines

Tissue and cell lysates were prepared, subjected to 10% SDS-polyacrylamide gel electrophoresis (PAGE) and transferred to polyvinylidene fluoride (PVDF) membrane (Schleicher and Schuell, Dassel, Germany). Membranes were washed with tris-buffered saline (TBS) containing 0.1% Tween 20 (TBST), and then blocked for 1 h in TBST containing 5% skimmed milk. After washing the membranes with TBST, they were treated with appropriate antibodies against chemokines/chemokine receptors, (CCL19, CCL21, CCR5 and CCR7) and cytokines [interleukin (IL)-2, IL-4, IL-10 and interferon (IFN)- γ], diluted with TBST (1:1000) and incubated overnight at 4 °C. Membranes were washed and treated with HRP-conjugated goat anti-rat IgG (diluted to 1:3000–1:5000) (Santa Cruz Biotech) in TBST for 60 min. The protein bands were visualized using electrogenerated chemiluminescent (ECL) solution (Amersham Biosciences UK Limited, little Chalfont, Buckinghamshire, UK).

Analysis of cytokines production

To assess the amount of IL-4, IL-10, IL-12 and IFN- γ in serum from the different experimental groups on days 7, 20 and 30 after transplantation, the levels of these cytokines were recorded using ELISA according to the manufacturer's instructions (BD Bioscience, San Jose, CA, USA).

FACS analysis

The cell surface expression of proteins on DCs was analysed by using flow cytometry at day 7 post-transplantation. Single cell suspensions were prepared from spleens and positively selected by using DC surface markers CD11c, MHC class II (H-A^d). Co-stimulatory molecule including CD40, CD80, CD86, CD83 and OX40L were analysed.

The levels of CD4⁺CD25⁺Foxp3⁺ Tr cells in different groups after transplantation were analyzed by flow cytometry. According manufacture's protocol, cells were stained for CD4 and CD25 for 30 min at 4 °C. After extracellular staining cells were permeabilized and stained for Foxp3. Appropriate isotype-matched IgGs were used as negative controls.

T-cell proliferation assays

Single cell suspensions were prepared from the spleens of the different groups (BALB/c) at day 7 post-transplantation and were co-cultured with mitomycin C treated C57L/B6 spleen cells or 129 spleen cells as stimulator cells for 48 h at 37 °C. For the final 18 h, individual wells were

pulse-labelled with 1 μ Ci of [³H] thymidine. The amount of radioisotope incorporated was determined using liquid scintillation counting.

Measurement of pathogen sensing receptors

cDNAs encoding mDC-SIGN and TLR4 were determined by reverse transcription-PCR (RT-PCR) using RNA isolated from bone marrow derived immature DCs treated with TPT and/or Rapa as a template. The primers of DC-SIGN are: F1-GCACTGAGAAGTGGCTGTGA and R1-CTTGCTAGGGCAGGAAGTTG; the primers for TLR4 are: F1-GCATGGCTTACACCACCTCT and R1-CAGGCTGTTTGTTCCCAAAT. The protein levels of these two receptors in DCs were analysed by Western blotting and flow cytometry.

Statistical analysis

All analyses for statistically significant differences were performed using the Student paired *t*-test. *P* < 0.05 was considered significant. All cultures were performed in triplicate, and error bars represent the standard deviation (SD). Kaplan–Meier method and the log-rank test were performed by SPSS software (SPSS Inc., Chicago, IL, USA) for animal survival in different treatment groups.

Results

TPT/Rapa treatment prolongs cardiac allograft survival

To determine the influence of TPT and Rapa alone or in their combination on allograft survival, heterotopic heart transplantations were established and the recipients received different immunosuppressant treatment and were monitored for graft survival. Untreated heart grafts were acutely rejected [mean survival time (MST): 7.66 ± 0.8 days, *n* = 6], whereas Rapa treatment alone prolonged the graft survival (MST: 22 ± 1.3 days, *n* = 6, *P* < 0.001 compared to untreated group). Similarly in group receiving only TPT the MST: 23.5 ± 5.3 days, *n* = 6 (*P* < 0.001). More significantly, in BALB/C recipients treated with the combination of TPT (3 mg/kg/day) and Rapa (1 mg/kg/day) there was an additive prolongation of graft survival (MST: 93.5 ± 6.7 days, *n* = 6), compared to the untreated recipients (*P* < 0.001) or those animals treated with either TPT or Rapa alone (Fig. 1).

Histological assessment of cardiac allograft

The histopathological changes were assessed in H&E stained sections at days 7 and 9 post-transplantation. Tissue necrosis, inflammatory cell infiltration observed in the untreated animals was reduced in recipients treated

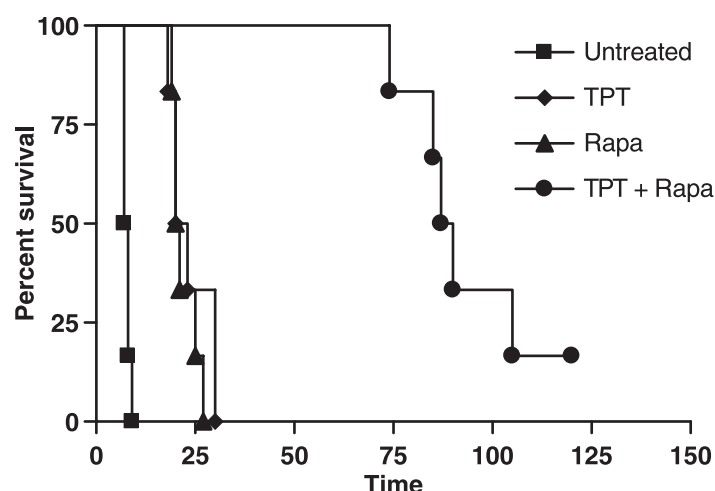


Figure 1 TPT/Rapa combined therapy prolongs allograft survival. BALB/c mice were transplanted with C57Bl/6 cardiac allograft and received different immunosuppressive treatments (days 0–7, 9, 11, 13, 15, $n = 6$ in each group) post-transplantation. MST for each treatment group was counted. The MST for TPT group is 23.5 ± 5.3 days and for Rapa group is 22 ± 1.3 days. The combination treatment significantly prolong allograft survival (MST: 93.5 ± 6.7 days). The MST is 7.7 ± 0.8 days in untreated group. $^{**}P < 0.001$ when compared TPT, Rapa and combination groups versus untreated group.

with TPT or Rapa alone. This reduction in pathology marked more in the TPT/Rapa in group (Fig. 2a). The cellular infiltration was further investigated by specific staining specific for T cells (CD3), macrophages (F4/80) and DCs (33D1). The number of each cell type was significantly reduced (ca 30%) in the groups treated with TPT and Rapa alone ($P < 0.05$) and even more so in combined therapy group ($>50\%$, $P < 0.01$) (Fig. 2a and b).

CD4⁺CD25⁺ T cells are expanded in TPT/Rapa-treated mice post-transplantation

In order to determine if TPT and Rapa treatment resulted in the expansion of a population of Tr cells, the expression of CD25, Foxp3 and CD127 was examined 7 days after transplantation in splenic CD4⁺ T cells. The results revealed that a population of T cells expressing CD4, CD25 and Foxp3 was significantly expanded (70.33% vs. 1.72%, $P < 0.05$) in spleen of recipients of TPT/Rapa treatment as compared to untreated mice (Fig. 3a). As regards CD127, which is not expressed on Tr cells [15] levels of expression were lower in the TPT/Rapa treatment group than in the controls (4.52% vs. 18.77%, $P < 0.01$, Fig. 3b).

Furthermore, we observed that BALB/c splenocytes isolated on day 7 after transplantation from the TPT/Rapa group had a reduced proliferative response when stimulated with mitomycin C treated C57Bl/6 spleen cells as compared to the control groups (Fig. 3c). In order to determine if these BALB/c splenocytes were tolerant to specific antigen they were cultured with 129 mice spleen cells as third-party stimulators. The results revealed that the proliferative response of the BALB/c splenocytes from TPT and/or Rapa groups was not affected when third-party stimulator spleen cells were used (Fig. 3d).

TPT/Rapa treatment decreases the expression of the chemokine receptors/ligands CCR7, CCR5, CCL19 and CCL21

Expression of lymph node-homing chemokine receptors CCR5 and CCR7 and their ligands CCL19 and CCL21 on DCs determine their migration to lymphoid tissues during priming [16,17], while effector T cells primarily use CCR5 to traffic to allografts during rejection [18]. We noted that both CCR5 and CCR7 were expressed at significantly lower levels in spleens isolated on day 7 after transplantation in the TPT/Rapa group compared to the controls. Furthermore, CCL19 but not CCL21

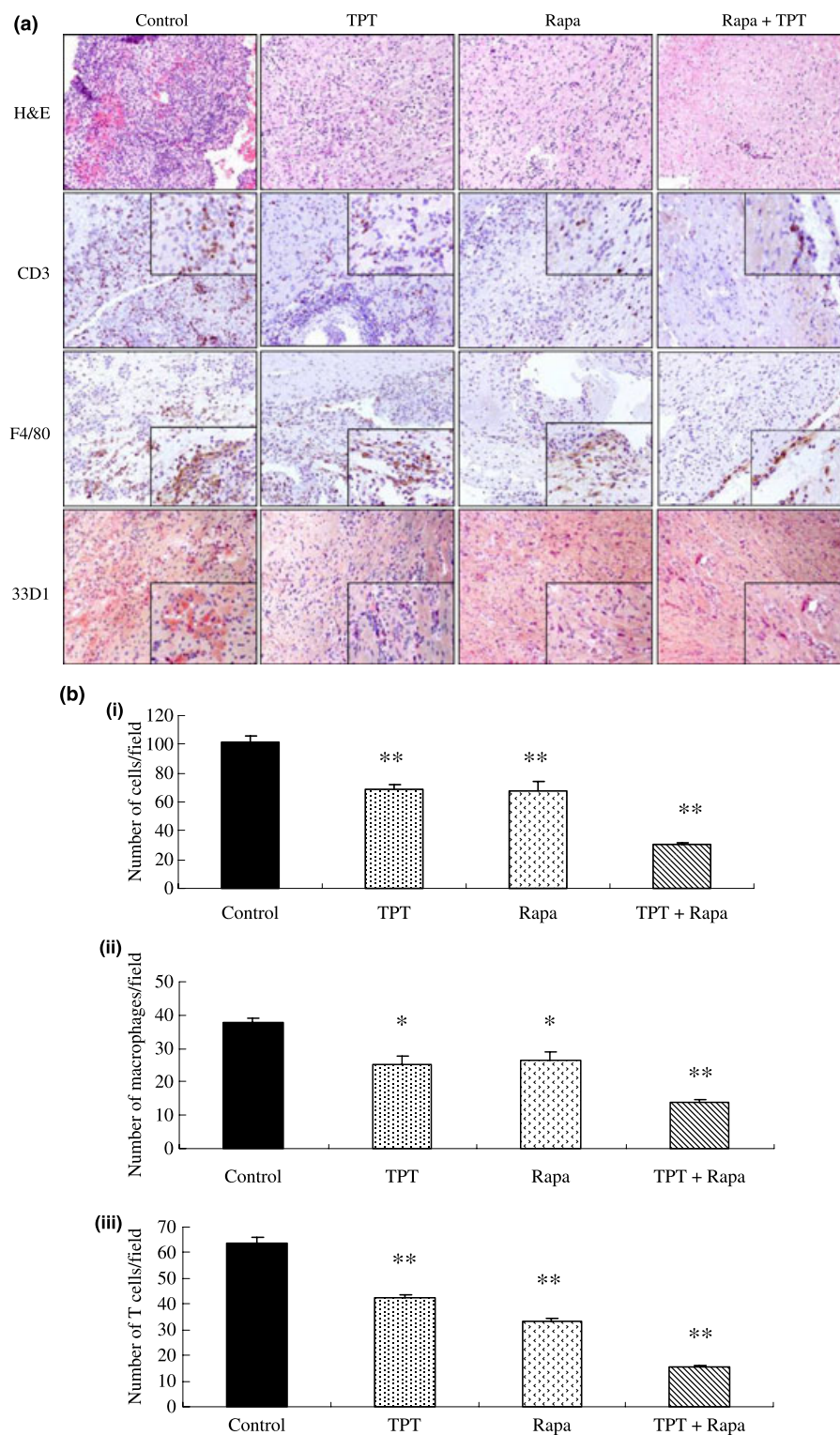


Figure 2 Histological assessment of cardiac allografts. (a) The histopathological changes were assessed in H&E and immunohistochemically stained sections. For T cell, macrophage and DC infiltration antibodies for CD3, F4/80 and 33D1 respectively were used. The positive signals were developed using DAB and counterstained with haematoxylin. (b) Quantitative analysis of cell infiltration with and without TPT and/or Rapa treatment. The total inflammatory cell infiltrate (i) and that of macrophages (ii), and T cells (iii) in the allografts was analysed 7 days after transplantation. * $P < 0.05$, ** $P < 0.001$ are compared with control group.

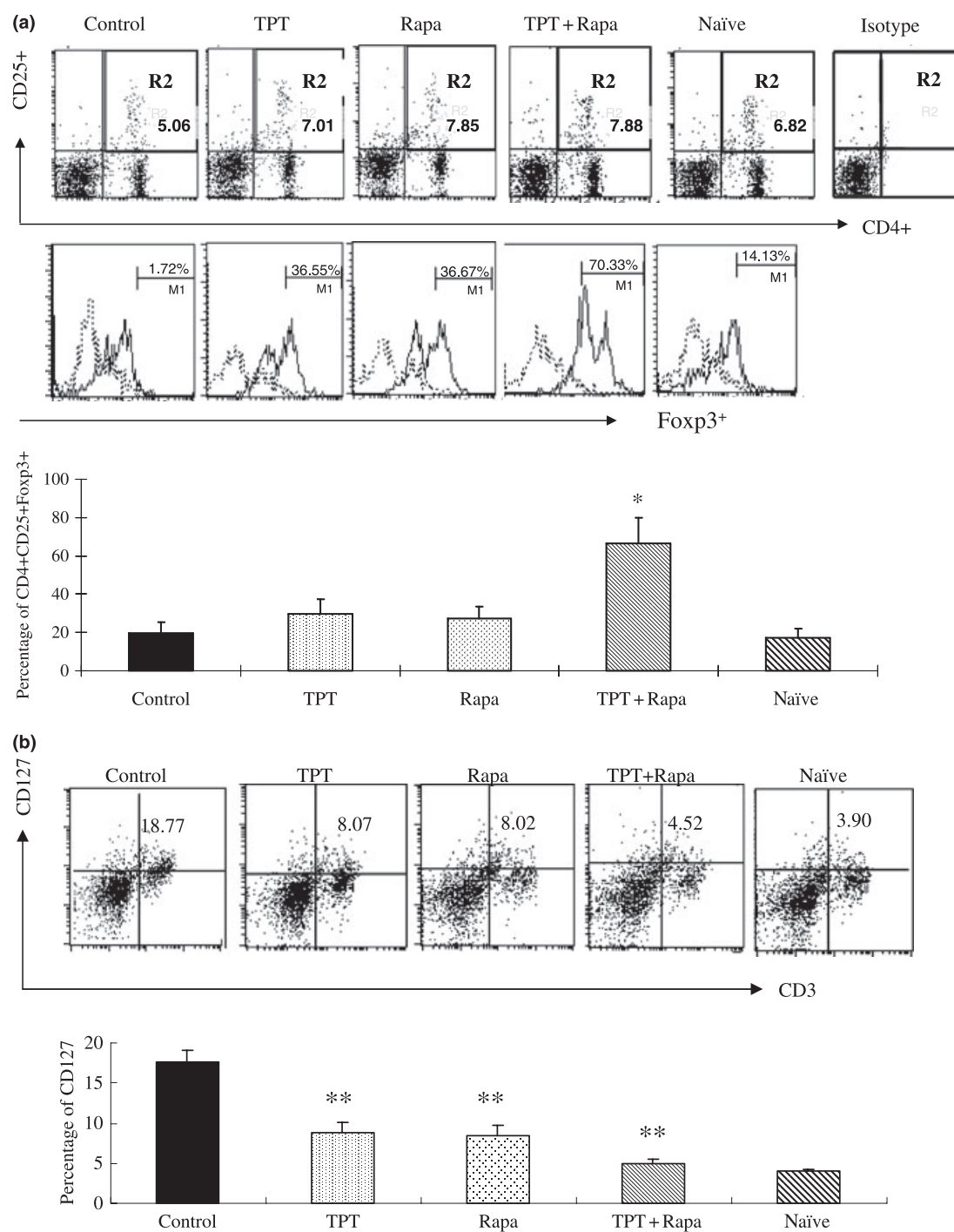


Figure 3 Effect of TPT/Rapa in expansion of Tr cells population in recipient's spleen after transplantation. (a) After CD4 and CD25 positive cells were gated at R2, Foxp3 high expressed cells were counted. The population of T cells expressing of CD4, CD25 and Foxp3 was compared in control group (without treatment), TPT- and/or Rapa-treated group and naïve group (syngenic transplantation group). * $P < 0.05$ when compared between TPT + Rapa to control group. (b) After CD4 and CD25 positive cells were gated, CD127 expressed cells were counted and compared in the different treated group. ** $P < 0.01$ when compared between treatment groups (TPT and/or Rapa) to control group. (c) BALB/c spleen cells from the different groups were isolated on day 7 post-transplantation and stimulated with mitomycin C treated donor C57Bl/6 spleen cells in MLRs. For naïve group, spleen cells were stimulated with donor BALB/c spleen cells. The results are representative of three independent experiments. ** $P < 0.01$ compared to control group. (d) BALB/c spleen cells from the different groups were isolated on day 7 post-transplantation and stimulated with mitomycin C treated the 129 spleen cells in MLRs. The results are representative of three independent experiments.

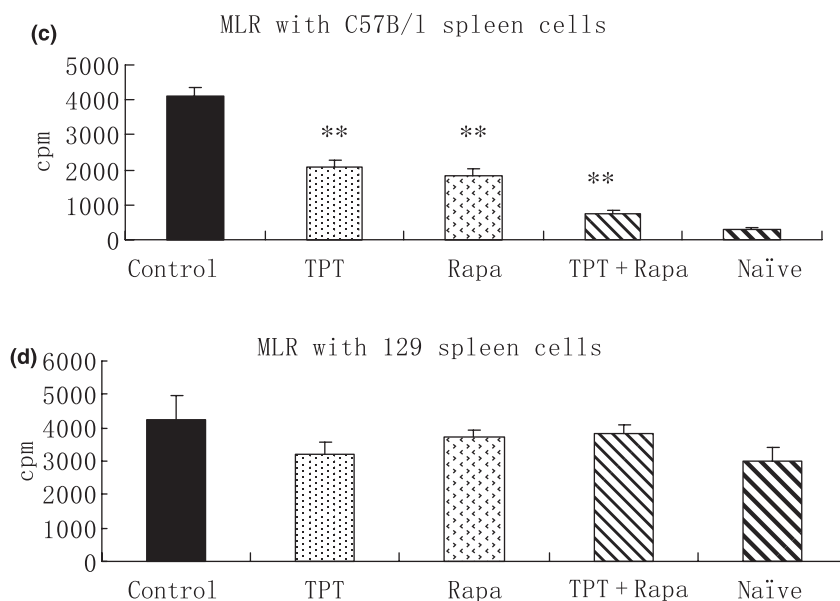


Figure 3 continued

was also markedly down-regulated with the combination treatment (Fig. 4a).

TPT/Rapa treatment promotes IL-4 and IL-10 production

Levels of Th1/Th2 cytokines were analyzed in spleens of allograft recipients at day 7 by Western blotting and the serum levels of these cytokines at days 7, 20 and 30 by ELISA. TPT/Rapa treatment up-regulated the expression of IL-4 and IL-10 and reduced the production of IL-12 and IFN- γ in the spleen (Fig. 4b). Likewise in the serum, TPT/Rapa also enhanced the levels of IL-4 and IL-10 which were maintained even at day 30 after transplantation. This was accompanied by a reduction in the levels of IL-12 and IFN- γ (Fig. 4c).

TPT/Rapa treatment inhibits DC maturation

Both Rapa and TPT can inhibit DC maturation, pro-inflammatory cytokine production and their ability to stimulate alloreactive T cells [19]. Here we analyze the effects of TPT/Rapa treatment on the cell surface phenotype of DCs isolated on day 7 after transplantation. DCs exposed *in vivo* to TPT/Rapa expressed markedly lower levels of CD83 compared to cells from either untreated animals or the single-treatment groups. Other co-stimulatory molecules, including CD40, CD80 and CD86, were also decreased in TPT/Rapa group (Fig. 5a). OX40L, which is expressed by mature DCs and contributes to co-stimulatory activity, was down-regulated in all treatment groups (Fig. 5b).

TPT/Rapa-treated modulates the expression of the pathogen-sensing receptors DC-SIGN and TLR4

Using both Western blotting and RT-PCR, we investigated the expression of TLR4 and DC-SIGN in splenic DCs from the different treatment groups on day 7 after transplantation (Fig. 6). The *in vivo* administration of these two compounds either alone or in combination down-regulated TLR4. As regards DC-SIGN, both protein and specific transcript levels were increased in the TPT- and TPT/Rapa-treated groups but not in those animals receiving Rapa alone.

Discussion

Immunosuppression is necessary to prevent rejection of allografts and current strategies are directed towards combination therapy using different immunosuppressive agents in an attempt to reduce adverse side effects [20]. Both TPT and Rapa have been studied independently as immunosuppressive drugs in experimental models of transplantation [12], but their combined effects on graft survival has not previously been investigated, which forms the focus of this study. Here, we report that combination treatment with TPT and Rapa increases the survival of fully MHC-mismatched allografts in a mouse model of cardiac transplantation by modulating key aspects of immunity. Graft survival is prolonged greater than four-fold than that which occurs when the recipients are treated with TPT or Rapa alone. Enhanced graft survival is not simply a concentration effect since increasing dosage

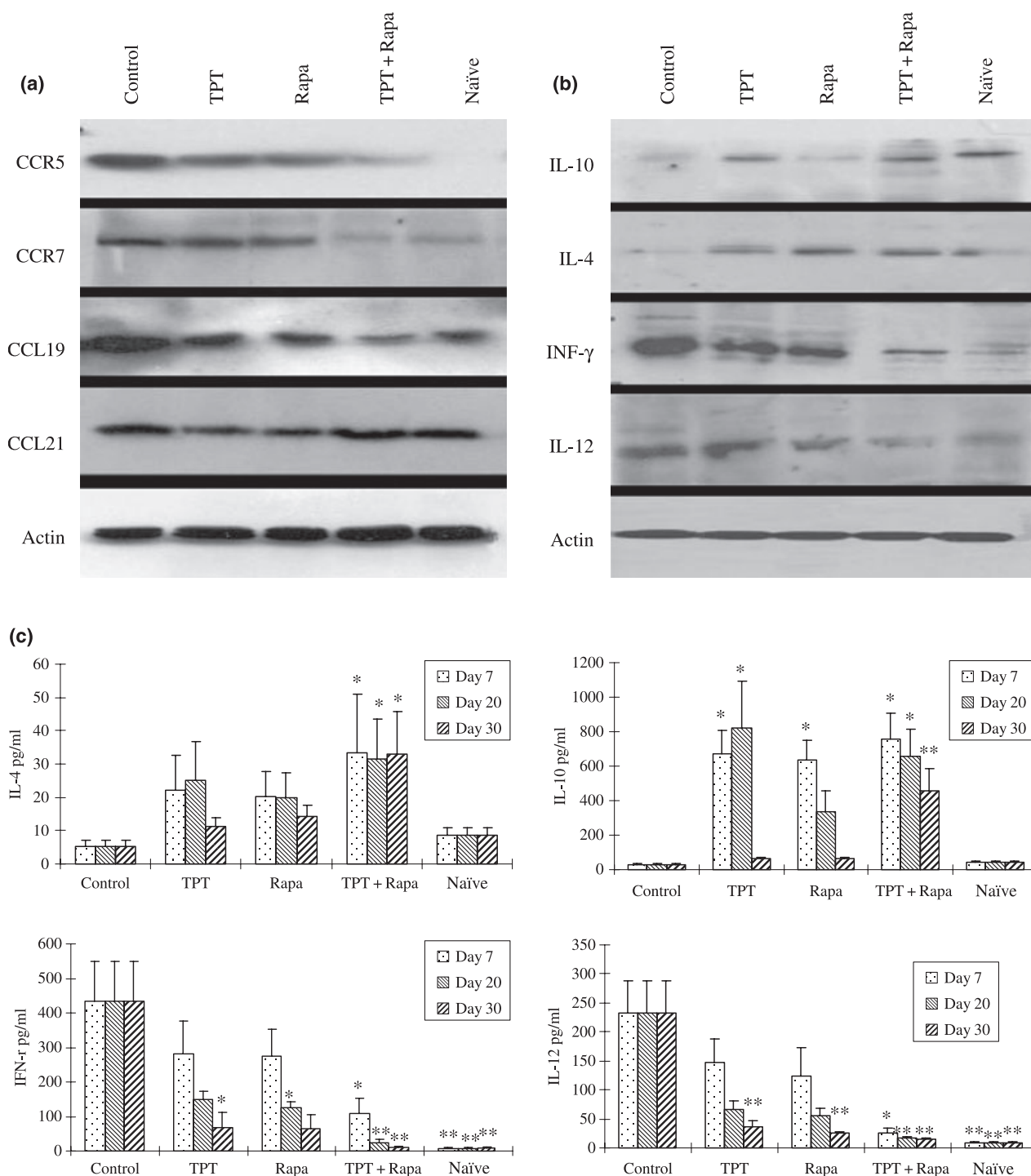


Figure 4 The expression of chemokine receptors/ligands (CCR7, CCR5, CCL19 and CCL21) and cytokines (Th1 and Th2 type cytokines). On day 7 post-transplantation spleen cell lysates were prepared from different experimental groups. CCL19, CCL21, CCR5 and CCR7 (a); and cytokines IL-4, IL-10, IL-12 and IFN- γ (b) were analysed by Western blotting. Levels of IL-4, IL-10, IL-12 and IFN- γ in the serum of different groups were measured by ELISA on days 7, 20 and 30 after transplantation (c). The results are representative of three independent experiments.

of either drug did not prolong survival time longer (data not shown). In agreement with others [7,21] we observe that TPT or Rapa reduces the severity of tissue damage

and the inflammatory cell infiltrate but when administered together again the reduction in the pathology is more notable.

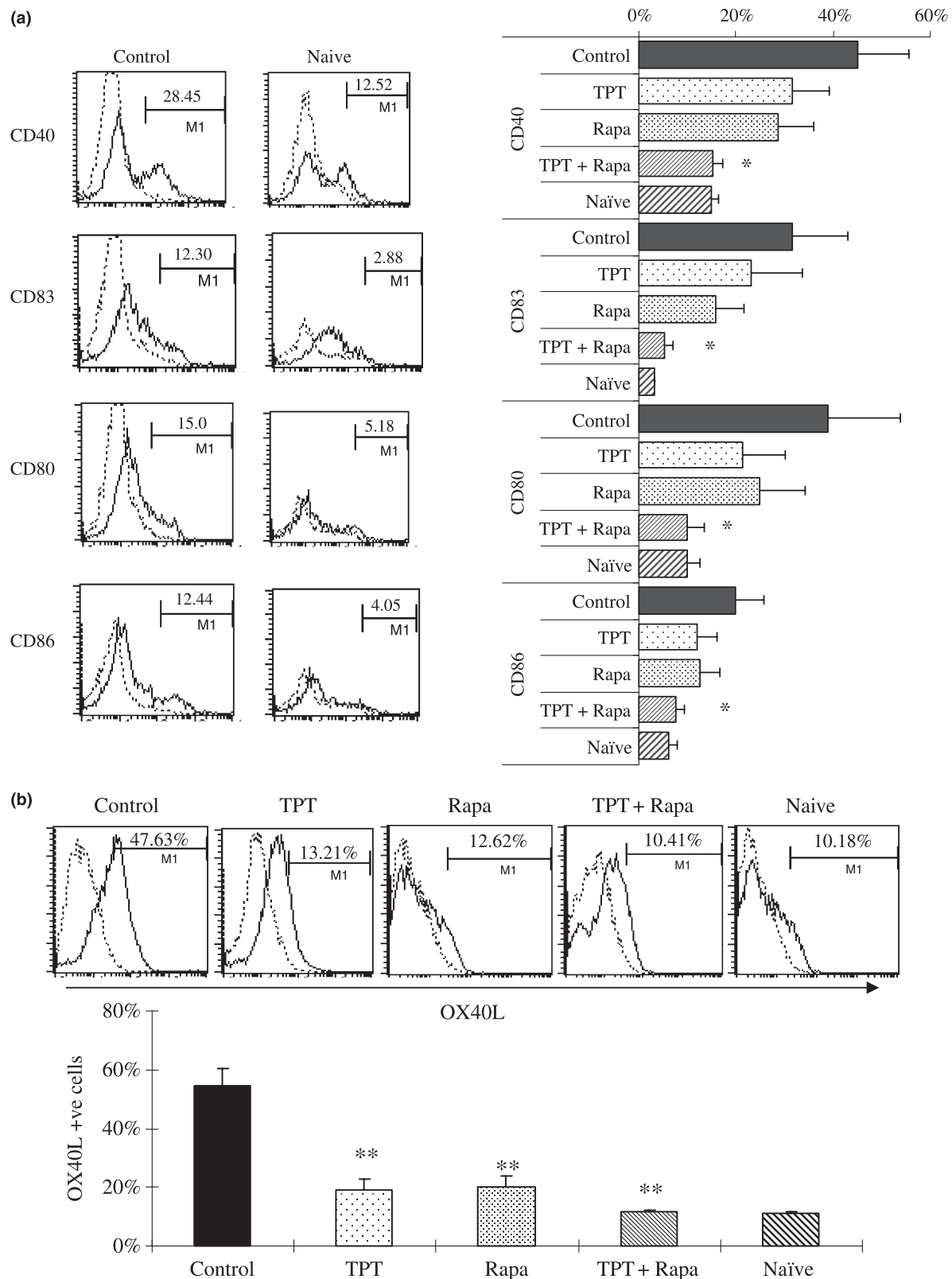


Figure 5 TPT/Rapa inhibits DC maturation. Expression of the co-stimulatory molecules (CD40, 80, 83 and 86) (a) and OX40L (b) were measured on splenic DCs at day 7 by flow cytometry. The results are representative of three independent experiments.

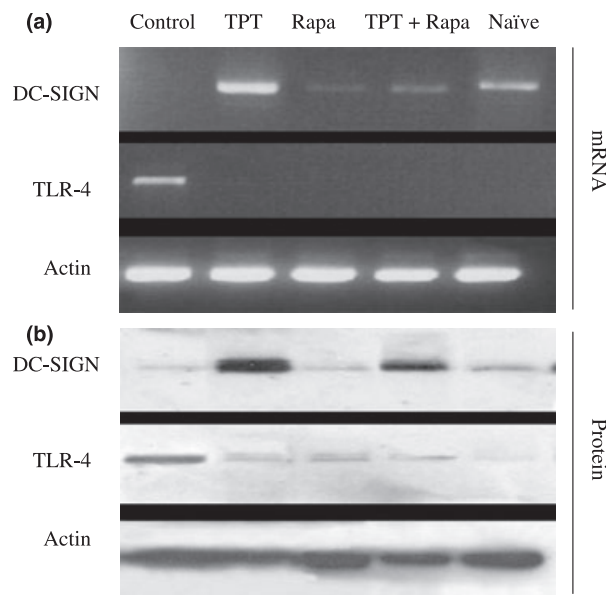


Figure 6 Effect of TPT/Rapa treatment on the expression of TLR4 and DC-SIGN. On day 7 post-transplantation spleen cells were isolated from the different experimental groups. Specific mRNA transcripts for DC-SIGN and TLR4 mRNA level were analyzed using RT-PCR (a). DC-SIGN and TLR4 protein expression was determined by Western blotting (b).

There is a large body of evidence that a population of Tr cells ($CD4^+CD25^+$) are induced or expanded in many experimental models of transplantation tolerance [22]. We too observe that this cell population is expanded and is antigen-specific following treatment with either TPT or Rapa alone but the combination therapy results in a marked increase in the percentage of $CD4^+CD25^+$ T cells expressing Foxp3. It is documented that Foxp3 is a transcription factor that confers regulatory function in nTr cells [23]. Expression of CD127 discriminates $CD127^{low}$ Tr cells from $CD127^{high}$ conventional T cells [24] and similarly the $CD4^+CD25^+Foxp3^+$ T cells present in the TPT/Rapa-treated graft recipients having low expression of CD127.

Exposure to TPT *in vitro* can modulate the surface expression of chemokine receptors and their ligands on DCs [6] as well as their ability to synthesize cytokines [13]. Splenic DCs from recipients treated *in vivo* with TPT/Rapa had reduced expression of CCR5, CCR7 and their ligand CCL19 was also down-regulated but no changes in the level of CCL21 were detected. The chemokines CCL19 and CCL21 are important in directing DCs to lymphoid tissues during antigen priming [14,25]. The low level of expression of these chemokine receptors and CCL19 suggests that the initiation phase of adaptive immunity induced by the allograft is less vigorous. As the expression of CCL21 was not altered in any of the treat-

ment implies that CCL19 plays a more important role in DC trafficking in this model. The ability of TPT and Rapa, given as individual monotherapy to modulate synthesis of Th1 and Th2 cytokine production has been investigated [26,27]. Here, we extend those findings to graft recipients treated with both compounds together and compare cytokine levels in lymphoid tissues and the serum. IL-4 and IL-10 expression was increased in both the spleen and serum and was accompanied by a reduction in the Th1 cytokines (IFN- γ and IL-12). Immune reactivity induced by the allografts is mostly of the Th1-type mediated and reciprocal regulation of Th1- and Th2 immunity [18]. There the ability of the TPT/Rapa to shift the balance towards a Th2 phenotype would reduce the pro-inflammatory nature of the alloresponse. IL-10 conditions the antigen presenting cells to adopt a tolerogenic phenotype and promote the induction of Tr cells [28], which in part would explain the increase in $CD4^+CD25^+Foxp3^+$ T cells that were detected in the TPT/Rapa-treated recipients.

Combination treatment with TPT/Rapa modulates both the cell surface and functional phenotype of DCs. Our previous *in vitro* studies have revealed that TPT-treated DCs have an immature phenotype and promotes the expansion of $CD4^+CD25^+Foxp3^+$ T cells [24]. It has recently been reported that Rapa [29] has similar effects. In addition to the down-regulation of CD40, CD80, CD83 and CD86 on the cell surface of DCs *ex vivo* from the TPT/Rapa-treated recipients; we also observed that the expression of OX40L was reduced. OX40/OX40L interactions are associated with T-cell activation and migration [30] and interrupting their binding has been reported to prolong graft survival [31]. The co-stimulatory activity of OX40 ligation blocks $Foxp3^+$ Tr cells [32] and therefore a lower expression of OX40L would reduce this function and facilitate the expansion of Tr cells. DCs from the TPT/Rapa-treated recipients have reduced T-cell stimulatory capacity which has been noted by others [33].

Dendritic cells express a number of pathogen-sensing receptors, which include TLR4 and DC-SIGN [34]. Ligation of TLR4 induces pro-inflammatory activity in DCs whereas DC-SIGN mediates immunosuppressive signals and it appears that there is cross-talk between the two families of receptors [34]. TPT and Rapa either alone or in combination inhibit the expression induced following graft transplantation. However, their effect on DC-SIGN differed in that TPT but not Rapa-induced expression in graft recipients and induction was observed in mice treated with TPT and Rapa. Increased expression of DC-SIGN would be expected to promote suppressive phenotype in the DCs, which would favour graft survival. With the exception of this finding, it is also important to note that the biological activity of TPT is additive or

synergistic with that of Rapa for the immunological parameters that we have investigated here and there is no evidence of antagonism.

Although the combination of TPT and Rapa significantly prolongs the grafts survival, the number of grafts viable at 120 days was still low (1/6). Histopathological analysis indicated that there were arteriosclerotic changes in the rejected grafts (data not shown), which suggests the current treatment regimen is active principally in preventing acute rejection and less effective in treating CR. However, it may be possible that further modifications in the treatment protocols may increase the efficacy of TPT in CR and increased knowledge of the mechanisms of action of TPT, which at present are limited, will aid in their design. Recent studies by Leuenroth *et al.* [35] have provided some information on the molecular mechanisms of TPT. Using an *in vitro* culture system of mouse kidney epithelial cells, they demonstrated that TPT binds to the calcium (Ca^{2+}) channel polycystin-2 (PC2) [35] and the TPT-binding activity was largely influenced by extracellular calcium concentration [36]. Ongoing studies in our group have revealed that TPT activity does not target NFATc1, unlike FK506 (data not shown). Further studies on the molecular pathways targeted by TPT will guide its clinical application.

In present study, we demonstrate that TPT and Rapa used in combination significantly prolongs allograft survival in a mouse model of cardiac transplantation. This was accompanied by the improved histological appearance and expansion of Tr cells in the recipients after transplantation. Furthermore, these Tr cells are able to mediate antigen-specific tolerance. These compounds, when administered together, attenuated rejection by inhibiting DC maturation and reducing T-cell reactivity. They also reprogrammed the chemokine receptor/ligand expression and the balance of Th1 and Th2 in graft-recipients towards a homeostatic phenotype. Finally, our results support the increased interest in combined immunosuppressive drug therapy in the management of graft rejection.

Acknowledgement

This work is supported by Hong Kong Competitive Earmarked Research Grant (HKU/7349/02M) from Hong Kong Research Grant Council. The authors have no competing interests that might be perceived to influence the results and discussion reported in this paper.

Authorship

YL and YC: designed, performed research and paper writing; FQL: performed research; JRL and PKHT: data analysis and paper writing.

References

- Ozdemir BH, Ozdemir FN, Demirhan B, Haberal M. TGF- β 1 expression in renal allograft rejection and cyclosporine A toxicity. *Transplantation* 2005; **80**: 1681.
- Yu DY. Clinical observation of 144 cases of rheumatoid arthritis treated with glycoside of Radix Tripterygium Wilfordii. *J Tradit Chin Med* 1983; **3**: 125.
- Tao XL, Sun Y, Dong Y, *et al.* A prospective, controlled, double-blind, cross-over study of tripterygium wilfordii hook F in treatment of rheumatoid arthritis. *Chin Med J (Engl)* 1989; **102**: 327.
- Ao JH, Li YT, Xiao XR. [Clinical study on the use of multiglycosides of Tripterygium wilfordii after cadaveric kidney transplantation]. *Zhonghua Wai Ke Za Zhi* 1994; **32**: 175.
- Tao X, Lipsky PE. The Chinese anti-inflammatory and immunosuppressive herbal remedy Tripterygium wilfordii Hook F. *Rheum Dis Clin North Am* 2000; **26**: 29, viii.
- Chen X, Murakami T, Oppenheim JJ, Howard OM. Triptolide, a constituent of immunosuppressive Chinese herbal medicine, is a potent suppressor of dendritic-cell maturation and trafficking. *Blood* 2005; **106**: 2409.
- Saunders RN, Metcalfe MS, Nicholson ML. Rapamycin in transplantation: a review of the evidence. *Kidney Int* 2001; **59**: 3.
- Hackstein H, Thomson AW. Dendritic cells: emerging pharmacological targets of immunosuppressive drugs. *Nat Rev Immunol* 2004; **4**: 24.
- Li R, Takazawa K, Suzuki H, *et al.* Synergistic effect of triptolide and tacrolimus on rat cardiac allotransplantation. *Jpn Heart J* 2004; **45**: 657.
- Wang J, Xu R, Jin R, Chen Z, Fidler JM. Immunosuppressive activity of the Chinese medicinal plant Tripterygium wilfordii. I. Prolongation of rat cardiac and renal allograft survival by the PG27 extract and immunosuppressive synergy in combination therapy with cyclosporine. *Transplantation* 2000; **70**: 447.
- Ho S, Clipstone N, Timmermann L, *et al.* The mechanism of action of cyclosporin A and FK5. *Clin Immunol Immunopathol* 1996; **80**: S40.
- Kingsley CI, Nadig SN, Wood KJ. Transplantation tolerance: lessons from experimental rodent models. *Transpl Int* 2007; **20**: 828.
- Liu Y, Chen Y, Lamb JR, Tam P. Triptolide, a component of chinese herbal medicine, modulates the functional phenotype of dendritic cells. *Transplantation* 2007; **84**: 1517.
- Eng CP, Gullo-Brown J, Chang JY, Sehgal SN. Inhibition of skin graft rejection in mice by rapamycin: a novel immunosuppressive macrolide. *Transplant Proc* 1991; **23**: 868.
- Banham AH. Cell-surface IL-7 receptor expression facilitates the purification of FOXP3(+) regulatory T cells. *Trends Immunol* 2006; **27**: 541.

16. Randolph GJ, Angeli V, Swartz MA. Dendritic-cell trafficking to lymph nodes through lymphatic vessels. *Nat Rev Immunol* 2005; **5**: 617.
17. Randolph GJ. Dendritic cell migration to lymph nodes: cytokines, chemokines, and lipid mediators. *Semin Immunol* 2001; **13**: 267.
18. Oppermann M. Chemokine receptor CCR5: insights into structure, function, and regulation. *Cell Signal* 2004; **16**: 1201.
19. Zhai Y, Ghobrial RM, Busuttill RW, Kupiec-Weglinski JW. Th1 and Th2 cytokines in organ transplantation: paradigm lost? *Crit Rev Immunol* 1999; **19**: 155.
20. Turvey SE, Wood KJ. Immunobiology of solid organ transplantation. *Int Surg* 1999; **84**: 279.
21. Qiu D, Kao PN. Immunosuppressive and anti-inflammatory mechanisms of triptolide, the principal active diterpenoid from the Chinese medicinal herb Tripterygium wilfordii Hook. f. *Drugs R D* 2003; **4**: 1.
22. Groux H, O'Garra A, Bigler M, et al. A CD4+ T-cell subset inhibits antigen-specific T-cell responses and prevents colitis. *Nature* 1997; **389**: 737.
23. Sakaguchi S, Ono M, Setoguchi R, et al. Foxp3+ CD25+ CD4+ natural regulatory T cells in dominant self-tolerance and autoimmune disease. *Immunol Rev* 2006; **212**: 8.
24. Liu W, Putnam AL, Xu-Yu Z, et al. CD127 expression inversely correlates with FoxP3 and suppressive function of human CD4+ T reg cells. *J Exp Med* 2006; **203**: 1701.
25. Gunn MD, Kyuwa S, Tam C, et al. Mice lacking expression of secondary lymphoid organ chemokine have defects in lymphocyte homing and dendritic cell localization. *J Exp Med* 1999; **189**: 451.
26. Blazar BR, Taylor PA, Panoskaltsis-Mortari A, Vallera DA. Rapamycin inhibits the generation of graft-versus-host disease- and graft-versus-leukemia-causing T cells by interfering with the production of Th1 or Th1 cytotoxic cytokines. *J Immunol* 1998; **160**: 5355.
27. Lin KX, Wang CZ, Qian GS. [Effect of Tripterygium wilfordii on Th1, Th2 cytokines production in asthma patients]. *Zhongguo Zhong Xi Yi Jie He Za Zhi* 2001; **21**: 22.
28. Yamazaki S, Inaba K, Tarbell KV, Steinman RM. Dendritic cells expand antigen-specific Foxp3+ CD25+ CD4+ regulatory T cells including suppressors of alloreactivity. *Immunol Rev* 2006; **212**: 314.
29. Strauss L, Whiteside TL, Knights A, Bergmann C, Knuth A, Zippelius A. Selective survival of naturally occurring human CD4+CD25+Foxp3+ regulatory T cells cultured with rapamycin. *J Immunol* 2007; **178**: 320.
30. Salek-Ardakani S, Croft M. Regulation of CD4 T cell memory by OX40 (CD134). *Vaccine* 2006; **24**: 872.
31. Miura Y, Thoburn CJ, Bright EC, Arai S, Hess AD. Regulation of OX40 gene expression in graft-versus-host disease. *Transplant Proc* 2005; **37**: 57.
32. Vu MD, Xiao X, Gao W, et al. OX40 costimulation turns off Foxp3+ TREGS. *Blood* 2007; **110**: 2501.
33. Yates SF, Paterson AM, Nolan KF, et al. Induction of regulatory T cells and dominant tolerance by dendritic cells incapable of full activation. *J Immunol* 2007; **179**: 967.
34. Geijtenbeek TB, van Vliet SJ, Engering A, t Hart BA, van Kooyk Y. Self- and nonself-recognition by C-type lectins on dendritic cells. *Annu Rev Immunol* 2004; **22**: 33.
35. Leuenroth SJ, Okuhara D, Shotwell JD, et al. Triptolide is a traditional Chinese medicine-derived inhibitor of polycystic kidney disease. *Proc Natl Acad Sci U S A* 2007; **104**: 4389.
36. Leuenroth SJ, Crews CM. Studies on calcium dependence reveal multiple modes of action for triptolide. *Chem Biol* 2005; **12**: 1259.